INT 409: COMPUTER FORENSICS CASE STUDY

LAB ASSIGNMENT

INVESTIGATING RHION POSSESSION EVIDENCE

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EXECUTIVE SUMMARY

This report presents a structured digital forensic examination conducted in response to a fictional training investigation concerning suspected possession and possible transfer of restricted imagery. The investigation focuses on the examination of a supplied USB disk image, the identification and recovery of deleted or recoverable files, the assessment of possible concealed information within recovered media, and the analysis of associated File Transfer Protocol (FTP) and Hypertext Transfer Protocol (HTTP) network traffic. The primary objective of the examination was to determine what the supplied digital artefacts could establish concerning the existence, recovery, concealment, movement, and possible possession of relevant digital material. Particular attention was given to the relationship between files recovered from the USB disk image and objects reconstructed from the supplied network traces.

The examination was designed around the principles of forensic preservation, repeatability, integrity verification, and evidence traceability. Original course materials were to remain unchanged, while all investigative operations were conducted using verified working copies. Cryptographic hashing using the SHA-256 algorithm was adopted as the principal integrity-verification mechanism. The investigation covered five principal technical areas: disk-image and file-system examination, deleted-file recovery, metadata and steganography analysis, FTP session reconstruction, and HTTP traffic examination. Reconstructed objects were to be validated using their file signatures, structural characteristics, byte sizes, and cryptographic hashes wherever possible.

The interpretation of evidence was approached cautiously. The existence of a file within a recovered disk image does not automatically establish who created, downloaded, stored, or possessed it. Similarly, the presence of an FTP or HTTP transaction does not, without supporting evidence, establish that a particular individual initiated the activity or that a reconstructed object was successfully retained on another device.

OBJECTIVES AND SCOPE OF INVESTIGATION

The principal objectives of this investigation are:

1. To preserve the supplied digital evidence and establish its integrity through documented acquisition and SHA-256 hashing.
2. To examine the USB disk image and identify its partition structure, file system, directory records, and relevant allocated or unallocated areas.
3. To recover deleted or otherwise recoverable files using an authorised forensic recovery methodology.
4. To examine recovered media for relevant metadata and possible concealed information using the supplied steganography workflow.
5. To reconstruct and analyse material FTP sessions, including transfer direction, server responses, and transferred objects.
6. To examine the protected archive and establish its completeness, encryption characteristics, and relationship to other recovered artefacts where supported.

7. To analyse the supplied HTTP traffic and identify relevant requests, responses, endpoints, and reconstructable objects.
8. To correlate material findings into a chronological UTC timeline and determine the extent to which the evidence supports or limits the suspected activity.

Scope

The examination covers the following supplied evidence categories:

1. USB disk image.
2. FTP network trace identified as rhino.log.
3. HTTP network traces identified as rhino2.log and rhino3.log.
4. Recovered files and associated metadata.
5. Any protected archive reconstructed from the supplied evidence.
6. Supporting teaching materials and approved laboratory procedures.

The investigation does not extend to contacting external hosts, attempting access to recorded accounts, reusing recovered credentials, executing reconstructed binaries, or conducting examinations outside the authorised course environment.

EVIDENCE ACQUISITION, PRESERVATION AND INTEGRITY

The integrity of digital evidence is fundamental to a defensible forensic investigation. Any uncontrolled modification to an original evidence file may

compromise its evidential value or introduce uncertainty concerning the reliability of subsequent findings. For this reason, the examination follows a preservation-first approach. Original evidence files are retained in their supplied state, and investigative operations are performed only on designated working copies.

The following precautions form part of the required preservation procedure:

- Original evidence files are maintained separately from working files.
- Original filenames are recorded without unnecessary alteration.
- File sizes are documented in bytes.
- SHA-256 hashes are calculated for each original file.
- Working copies are created using a documented copying procedure.
- The SHA-256 values of original files and corresponding working copies are compared.
- Recovered files are stored separately from original evidence and working copies.
- All relevant commands, tool versions, and examination outputs are documented.
- Access to the evidence directory is restricted to the authorised examiner and approved laboratory environment.

No original course material is to be deleted, overwritten, reformatted, or otherwise altered during the investigation.

Evidence Identification Register

The following register identifies the principal evidence categories. Actual file sizes, hashes, and paths must be populated from the supplied evidence package.

Serial	Evidence ID	Original file name	Evidence description
1	E01	RHINOUSB.dd	Supplied USB disk image
2	E02	rhino.log	FTP network trace
3	E03	rhino2.log	HTTP network trace
4	E04	rhino3.log	HTTP network trace
5	E05	Prescription.pdf	Material recovered
6	E06		Reconstructed protected archive
7	E07	Receipt.pdf	Reconstructed object

Integrity Verification

SHA-256 hashing was selected to establish a reproducible integrity reference for each evidence file.

The hash of an original file is compared with the hash of its corresponding working copy. Where both values are identical, this supports the conclusion that the working copy is byte-for-byte identical to the original at the time of comparison.

Evidence Handling and Access Precautions

All evidence handling must remain within the approved course laboratory environment. The investigation must not expose potentially sensitive recovered material to unauthorised persons or external services.

Where screenshots are required, unrelated personal information, visible credentials, and sensitive imagery must be appropriately redacted. Redaction must not remove the technical details necessary to verify the finding, including relevant timestamps, file paths, packet references, stream numbers, and hash values.

FORENSIC METHODOLOGY

The investigation adopts a structured forensic methodology consisting of preservation, examination, recovery, validation, correlation, and reporting.

Each examination stage is documented independently so that an assessor can distinguish the original evidence, the method applied, the resulting artefact, and the interpretation made from it.

5.1 USB Disk-Image Examination

The USB disk image is the principal storage-media evidence source. The examination is intended to identify the image's logical structure and establish the file-system characteristics relevant to subsequent recovery operations. The Sleuth Kit suite provides the principal examination framework.

The following tools were used in the course of the investigation:

mmls: Examination of partition layout.

fsstat: Examination of file-system information.

fls: Listing of directory entries and file-system records.

icat: Extraction of file content using relevant metadata addresses.

The partition examination should identify the number of partitions, their starting sectors, ending sectors, sizes, and any unallocated areas represented in the image.

The file-system examination should identify the file-system type and relevant structural characteristics, including allocation information and metadata records.

Particular attention should be given to the distinction between allocated and unallocated storage. Allocated areas may contain records associated with currently referenced files, whereas unallocated areas may retain remnants of deleted file content or other recoverable data.

However, the presence of data in unallocated space does not automatically establish when a file was deleted, who deleted it, or whether the file was intentionally concealed.

Deleted-File Recovery

Deleted-file recovery is intended to identify files whose directory references or allocation records may no longer be active but whose contents remain recoverable.

The authorised course recovery method, such as PhotoRec or the demonstrated file-carving workflow, should be applied to a working copy of the disk image.

The following are the recovery process:

- The recovery tool and version.
- The source disk-image filename and hash.
- The selected recovery scope.
- The output directory.
- The file types or signatures targeted.
- The total number of recovered files.
- The names and sizes of material recovered files.
- SHA-256 hashes of relevant recovered files.
- File-type and metadata validation results.

Recovered files should be treated as separate evidence objects. Their recovery does not necessarily establish that they existed as intact files in their original directory structure. File carving may recover content without recovering the original filename, directory path, or complete file-system metadata. Fragmentation may also result in incomplete or corrupted output.

Duplicate detection should therefore be performed using hashes and relevant file characteristics. Files with identical SHA-256 values are byte-for-byte identical, while files with different hashes may still contain similar or related material.

Metadata Examination

Metadata examination provides additional information about recovered files and their recorded characteristics.

Relevant metadata may include:

- File format and extension.
- File size.
- Creation and modification timestamps, where available.
- Embedded application metadata.
- Image dimensions or media properties.
- Embedded comments or descriptive fields.
- File signatures and structural information.

The examination should distinguish between filesystem timestamps and metadata embedded within a file. These values may originate from different systems and may be affected by copying, extraction, conversion, or application-specific behaviour.

A timestamp was not treated as proof of the exact time a person viewed, created or transferred a file without appropriate corroboration.

Steganography Examination

Steganography involves concealing information within an apparently ordinary carrier file. In this investigation, the purpose of steganography

analysis is to determine whether any relevant recovered media contains indications of embedded information.

The examination should begin with ordinary file identification and metadata review before applying the supplied course steganography workflow.

The process:

- The identity and SHA-256 hash of the carrier file.
- Its actual file type, rather than relying solely on its extension.
- The metadata examination results.
- The steganography detection method and version.
- The detector output.
- Any justified extraction or recovery operation.
- The identity, size, and SHA-256 hash of any recovered embedded output.
- Any validation undertaken to establish whether the output is complete and readable.

A positive detection result should be interpreted carefully. Unusual file characteristics or embedded data do not automatically establish malicious concealment or illegal content.

Similarly, a negative result from a particular detection tool does not prove that a file contains no concealed information. Detection methods have technical limitations and may not identify every possible steganographic technique.

Any extraction of embedded material must remain within the approved course workflow. Potentially illegal imagery should not be opened, distributed, or uploaded for external examination.

FTP NETWORK-TRAFFIC EXAMINATION

Purpose of FTP Analysis

The FTP examination is intended to identify material file-transfer activity represented in the supplied rhino.log trace.

FTP commonly uses separate control and data connections. The control connection carries session commands and server responses, while the data connection carries transferred file content.

Consequently, a defensible examination must consider both the session-control records and the corresponding data-transfer records.

The presence of a login or file-transfer command alone is insufficient to establish that a complete file was successfully transferred.

Session Identification

The FTP capture was examined to identify:

- Client and server IP addresses.
- Relevant TCP ports.
- Session establishment and termination.
- FTP control-channel exchanges.
- Relevant commands and server responses.
- Data-channel establishment.
- Transfer direction.

- Object boundaries and reconstructed file content.

The investigation should preserve packet numbers and TCP stream references for each material session. The examiner should also distinguish between an upload and a download. The direction of transfer must be established from the protocol exchanges and the associated data connection rather than inferred from an isolated packet.

FTP Command and Response Interpretation

FTP commands may provide evidence of directory navigation, authentication attempts, file retrieval, file storage, or session termination. However, the evidential significance of each command depends on the corresponding server response and the data-channel activity. For example, a transfer-related command may indicate an attempt to retrieve or store an object. A corresponding successful server response and a complete data stream would provide stronger evidence that the transfer operation progressed.

Even then, the evidence must be interpreted in context. A complete network transfer does not independently establish that the receiving device retained the object permanently or that a named person controlled the session. Visible credentials must be treated as sensitive evidence. They should not be reused, tested, or unnecessarily reproduced in the report. Screenshots should redact credential values while preserving the relevant protocol context.

FTP Object Reconstruction

Where the supplied trace contains sufficient data, the relevant FTP object should be reconstructed using the approved Wireshark workflow.

The reconstruction record identify:

- The control-stream reference.
- The data-stream reference.
- Relevant packet ranges.
- The transfer direction.
- The reconstructed filename, where available.
- The resulting byte size.
- The reconstructed object's SHA-256 hash.
- The validation result.

The reconstructed file should be validated using file signatures, structural checks, and appropriate file-identification tools. A reconstructed object should not be assumed complete merely because an export operation produced a file. The available data must be checked for missing segments, incomplete captures, truncation, and protocol-related framing.

PROTECTED ARCHIVE EXAMINATION

Archive Identification

A protected archive reconstructed from the FTP or other supplied evidence must be examined to establish its identity, structural completeness, and encryption characteristics. The examination should record the archive filename, byte size, SHA-256 hash and source stream or recovery record.

The archive's file signature and internal structure should be examined to determine whether the reconstructed object is consistent with its claimed format.

7.2 Archive Completeness and Encryption

The archive examination should establish whether the file is structurally recognisable and whether its contents indicate encryption or password protection.

Evidence of encryption may include archive metadata, encryption flags, or the behaviour of the approved archive inspection tool.

An inability to list or extract archive contents does not, by itself, prove encryption. Corruption, truncation, unsupported compression methods, or incomplete reconstruction may produce similar results.

7.3 Authorised Password-Recovery Procedure

Where the lecturer has authorised a course-only password-recovery workflow, it may be documented as part of the examination of the supplied training archive. Any such activity must be limited strictly to that archive and the approved laboratory environment.

The report should identify the authorised tool, version, workflow, input archive hash, and outcome without exposing unnecessary credential

material. No recovered password should be reused against another file, account, host, or service.

The archive was successfully opened using the approved method, the extracted contents was treated as new evidence objects. Their filenames, sizes, hashes and extraction relationships was recorded. The hash of an extracted file may be compared with a corresponding recovered or network-reconstructed file. Identical hashes would establish binary equality between those examined objects, but would not independently establish their original source or the identity of the person responsible for their movement.

HTTP NETWORK-TRAFFIC EXAMINATION

Purpose of HTTP Analysis

The HTTP examination focuses on the supplied traces rhino2.log and rhino3.log. The objective is to identify relevant client requests, server responses, transferred objects, and any relationships between reconstructed network content and the files recovered from the USB disk image or FTP evidence. HTTP analysis must account for the relationship between request and response messages and the underlying TCP stream.

Request and Response Identification

The following Wireshark display filters provide appropriate starting points:

http

http.request

http.response

Relevant HTTP requests was examined:

Request method.

Requested resource or URI.

Host header.

Client endpoint.

Server endpoint.

Timestamp.

Associated TCP stream.

Relevant responses was also examined:

HTTP status code.

Content-Type header.

Content-Length, where available.

Transfer-Encoding, where applicable.

Response timestamp.

Associated TCP stream.

Availability of the complete response body.

A request does not necessarily establish that the requested object was successfully returned. The associated response must be examined to determine what the server actually supplied.

HTTP Object Reconstruction

Where the supplied capture contains a complete and reconstructable object, the approved Wireshark export or reconstruction workflow should was used.

The reconstructed object should be saved to a dedicated working directory and assigned a unique evidence ID.

CORRELATION OF DISK, RECOVERY, FTP AND HTTP EVIDENCE

Correlation Approach

The central analytical objective is to determine whether the independent evidence sources support a coherent sequence of events. The USB disk image, recovered files, metadata, FTP trace, and HTTP traces should not be examined as isolated collections. Their relationships should be evaluated using identifiable technical characteristics.

The principal correlation mechanisms include:

Cryptographic hash comparison: Determining whether recovered files and reconstructed network objects are byte-for-byte identical.

Timestamp comparison: Comparing recorded event times after documenting the relevant time zones and normalisation method.

Filename and file-type comparison: Assessing whether objects share meaningful names, extensions, or format characteristics.

File-size comparison: Identifying possible relationships between stored, recovered, and transferred objects.

Network-session correlation: Linking relevant requests, responses, and data streams to reconstructed objects.

File-system correlation: Comparing recovered content with directory records, metadata, and the storage locations from which it was obtained.

A strong correlation was ideally be supported by more than one independent characteristic.

For example, a matching SHA-256 hash between a network-reconstructed object and a file recovered from the USB image would establish that the two examined objects are byte-for-byte identical. If their associated timestamps and transfer records are also consistent, this would provide additional support for a relationship between the network activity and the stored object.

Distinguishing Observation from Inference

The report distinguishes between direct observations and interpretations.

Direct observation: A specific packet contains an HTTP request to a recorded resource.

Inference: The client may have attempted to retrieve the resource.

Direct observation: A reconstructed network object and a recovered disk-image file have identical SHA-256 hashes.

Inference: The two objects are byte-for-byte identical and may be related through a common transfer or storage process.

Direct observation: A file-system record identifies a recoverable file entry.

Inference: The file may previously have existed within the examined file system, subject to the limitations of the record and recovery method.

This distinction is essential because digital forensic evidence must support the conclusion drawn from it rather than merely appear consistent with a suspected explanation.

CORRELATED UTC TIMELINE

Time Normalisation

All material events was represented in Coordinated Universal Time (UTC) in the final timeline where timestamps are recorded in local time, the original time zone and the conversion applied was documented.

I also consider whether the timestamp originates from a network packet, file-system metadata, embedded file metadata, or a forensic tool's interpretation of a record. These sources may represent different event types and should not automatically be treated as interchangeable.

FINDINGS AND DISCUSSION

Evidence Preservation and Integrity

The preservation stage is intended to establish a reliable baseline for the examination. Each original evidence file was identified by its supplied

filename, source location, byte size, and SHA-256 hash. Its corresponding working copy must be independently hashed and compared with the original.

The preservation record should establish whether the examination was conducted on verified working copies and whether any integrity discrepancies were identified.

USB Disk-Image Structure

The USB disk image examination should establish its partition layout and file-system characteristics. The findings should identify the relevant partition or storage area, the file-system type, and the metadata records examined.

Where unallocated space was examined, the report should identify the examination scope and explain whether the records produced evidence of recoverable file content. The interpretation must account for the limitations of file-system metadata, particularly where deletion, fragmentation, or overwriting may affect the available records.

Deleted and Recoverable Files

The recovery findings should identify each material recovered file by its recovered filename, file type, size, SHA-256 hash, recovery method, and output location. The report should explain whether the recovered output was complete, whether its structure was validated, and whether duplicate files were identified. Where a recovered file lacks its original filename or directory information, this limitation must be explicitly stated.

A recovered file may demonstrate that content remained within the examined image and was recoverable at the time of examination. It does not independently establish the identity of the person who created, stored, or deleted it.

Hidden-Content Examination

The steganography examination should state whether the supplied detection workflow identified indications of embedded content in any relevant recovered media. A positive result should identify the carrier file, the detection method, the relevant output, and the validation undertaken. Where embedded content was recovered, its output filename, size, hash, and relationship to the carrier file should be documented. Where no relevant embedded content was detected, the report should state the method and scope of the examination rather than making an absolute claim that no hidden information exists.

FTP Findings

The FTP findings should identify each material session and the corresponding control and data-channel records. The report should explain the transfer direction, relevant commands, server responses, and whether the associated data stream contained a complete reconstructable object. Where an object was reconstructed, its hash and validation results should be included.

The distinction between an attempted transfer and a confirmed completed transfer was maintained throughout the analysis.

Protected Archive Findings

The archive examination should establish whether the reconstructed archive was complete and whether its internal characteristics supported an encryption interpretation. Where authorised password recovery was performed, the report should document the course-approved method and outcome without unnecessarily disclosing sensitive credentials.

Any extracted files should be independently hashed and compared with other relevant recovered or reconstructed objects.

HTTP Findings

The HTTP findings should identify relevant client requests and corresponding server responses in rhino2.log and rhino3.log. The report should document the endpoints, timestamps, request methods, response status codes, content types, packet numbers, and TCP streams associated with the material transactions.

Where an object was reconstructed, the report recorded its filename, size, hash and validation results. The interpretation was distinguished between a request, a server response, a complete object transfer, and evidence of subsequent storage.

Cross-Source Findings

The cross-source examination should identify whether any recovered disk-image files correspond to FTP or HTTP reconstructed objects. The strongest form of binary comparison is a matching SHA-256 hash. Other similarities,

including filenames, file sizes, timestamps, and file types, may provide supporting context but are not equivalent to cryptographic identity.

CONCLUSION

This investigation is designed to examine whether the supplied USB disk image and associated network traces provide mutually supporting evidence of relevant file storage, deletion, recovery, concealment, and network transfer activity. The examination combines file-system analysis, deleted-file recovery, metadata review, steganography testing, FTP session reconstruction, protected archive examination, HTTP analysis, and cryptographic comparison.

The principal strength of this approach is the ability to evaluate relationships across different evidence sources. A recovered file may establish the existence of recoverable content within the disk image, while a reconstructed network object may establish that corresponding binary content was present in a captured communication. If the two objects have identical SHA-256 hashes, that provides strong evidence that the examined objects are byte-for-byte identical.

However, the technical existence of a file, the recovery of its content, or the observation of a network transfer does not automatically establish knowing possession by a particular individual. The investigation must also distinguish between evidence of a transfer attempt and evidence of a completed transfer. FTP commands, HTTP requests, server responses, reconstructed objects, and local file-system records each have different evidential significance.

Accordingly, the final conclusion must be based on the actual verified examination results. Until the evidence registers, recovery records, network reconstructions, and cross-source comparisons have been completed, no definitive finding concerning illegal possession, successful transfer, concealment, or individual responsibility can be justified. The case requires a multi-source forensic interpretation. The strength of any conclusion will depend on the completeness, integrity, and consistency of the disk-image, recovered-media, FTP, and HTTP evidence.

LIMITATIONS OF THE INVESTIGATION

The following limitations must be considered when interpreting the findings.

Deleted-File Recovery Limitations

File carving may recover content without preserving the original filename, directory structure, or complete file-system metadata. Fragmentation, overwriting, and incomplete allocation records may also affect the accuracy and completeness of recovered files. Therefore, recovery of a file does not necessarily establish its original location or the circumstances of its deletion.

Incomplete Network Captures

A packet capture may contain only part of a communication session. Missing packets, truncated data, incomplete TCP streams, and capture interruptions may prevent complete reconstruction of transferred objects. A reconstructed object must therefore be validated before it is treated as a complete representation of the original transfer.

Encryption and Archive Accessibility

Encryption may prevent direct examination of archive contents. Even where an authorised course-only recovery method provides access, the result must be validated against the archive's integrity and structure. An inaccessible archive may also be corrupted or incomplete, so encryption should not be assumed without supporting technical evidence.

Timestamp Reliability

File-system, embedded metadata, and packet-capture timestamps may originate from different clocks and time zones. Clock differences, incorrect system settings, copying operations, and metadata modification can affect their interpretation. All timestamps should therefore be normalised to UTC where possible, with the original timestamp source and conversion method documented.

Possession and User Attribution

A file recovered from a USB image does not independently establish who physically possessed the USB device or who knew about the file. Likewise, an IP address, username, or network endpoint identifies a technical record or communication context, not necessarily the person responsible for the activity. Additional corroborating evidence would be required to support individual attribution.

Steganography Detection Limitations

Steganography detection tools may identify suspicious characteristics without establishing that meaningful information has been concealed. Conversely, a negative result cannot rule out all possible hiding techniques. Detection results must be evaluated alongside file structure, metadata, extraction output, and other relevant evidence.

Duplicate and Related Files

The existence of multiple files with similar names or content may result from copying, downloading, backup operations, or repeated recovery. Duplicate detection and hash comparison are necessary to prevent the same underlying object from being counted as several independent pieces of evidence.

FORENSIC RECOMMENDATIONS

Based on the requirements of the investigation, the following recommendations are appropriate:

1. Maintain the original evidence files unchanged and preserve their documented integrity records.
2. Ensure every material recovered or reconstructed object is assigned a unique evidence ID and SHA-256 hash.
3. Retain the relevant Sleuth Kit, recovery, steganography, and Wireshark outputs necessary to reproduce each material finding.

4. Validate network-reconstructed objects before using them as evidence of complete file transfers.
5. Correlate recovered media and network objects using hashes, timestamps, file characteristics, and identifiable source records.
6. Maintain a consistent UTC timeline and document any uncertainty in the underlying timestamps.
7. Avoid person-level attribution unless independent evidence supports a reliable connection between a technical event and a specific individual.
8. Preserve sensitive material in the approved laboratory environment and follow the lecturer's instructions concerning retention and eventual disposal.